chinooksalmonCompLtoe5CSilt model summary

Contents

Overview

This document summarizes the results of a global sensitivity and uncertainty analysis for the **chinook-salmonCompLtoe5CSilt** habitat suitability index (HSI) model for *Oncorhynchus tshawytscha*. Metadata for the model is stored in the ecorest package in R.

The original documentation for this model can be found here[1].

Sub-model: **Riverine compensatory limiting factor model for Chinook Salmon habitat with temperatures less than or equal to 5C during the egg and pre-emergent yolk sac fry period and fines less than or equal to 0.8mm in size (silt)**

The chinooksalmonCompLtoe5CSilt model is comprised of **19** variables and **3** components.

Variables:

Table 1. SIV variables included in the chinooksalmonCompLtoe5CSilt model. Type indicates whether a variable is continuous or categorical and breakpoints indicates the number of distinct breakpoints in suitability graphs.

	Variable name	Type	Breakpoints
SIV1	annual.max.or.min.pH.SIV	continuous	6
SIV2	max.temp.ad.warmest.SIV	continuous	5
SIV2B	max.temp.juv.warmest.SIV	continuous	5
SIV3	min.DO.ltoe5C.egg.dev.SIV	continuous	4
SIV4	pls.late.growing.season.SIV	continuous	4
SIV5	pls.class.late.growing.season.SIV	categorical	3
SIV6	max.min.temp.start.end.spwn.SIV	continuous	5
SIV7	max.min.temp.start.end.emb.incub.SIV	continuous	6
SIV8	spwn.gravel.pct.SIA.SIV	continuous	12
SIV8B	spwn.gravel.pct.SIB.SIV	continuous	10
SIV9	avg.wtr.column.vel.SIV	continuous	6
SIV10	avg.fines.silt.spwn.gravel.pct.SIV	continuous	11
SIV11	avg.annual.base.flow.late.summer.SIV	continuous	3
SIV12	multiples.of.avg.flow.SIV	continuous	15
SIV13	dominant.subs.rffl.run.SIV	categorical	3
SIV14	avg.fines.lt3mm.pct.SIV	continuous	9
SIV15	late.summer.N.levels.SIV	categorical	5
SIV16	strm.area.escape.cov.pct.SIV	continuous	3
SIV17	instream.area.10.to.40cm.SIV	continuous	3

[1] https://ecolibrary.sec.usace.army.mil/resource/75cc3ca3-1110-4ef6-fa99-142f99a846eb

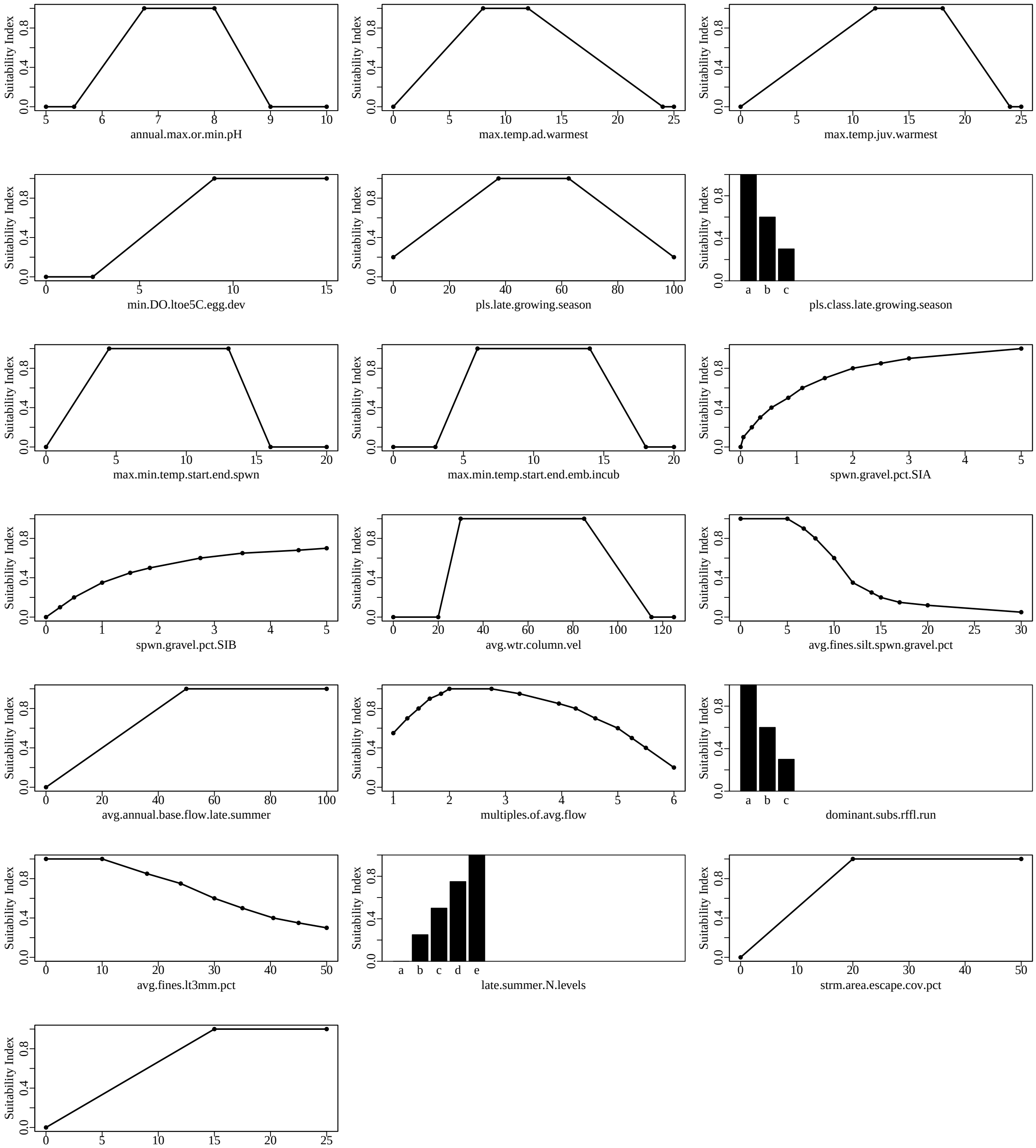

Figure 1. Suitability index graphs for variables included in the chinooksalmonCompLtoe5CSilt model in ecorest.

Components:

Table 2. Components included in the chinooksalmonCompLtoe5CSilt model in ecorest.

<table>
<tr><th>Component</th><th>Equation</th></tr>
<tr><td>Embryo component</td><td>ifelse(((SIV8+SIV8B)/2)<=0.3|SIV9<=0.3|SIV10<=0.3,min(SIV3,SIV6,SIV7,((SIV8+SIV8B)/2),SIV9,SIV10,SIV11,SIV12,na.rm=T),ifelse(((SIV8+SIV8B)/2)==SIV9 & SIV9==SIV10,min(SIV3,SIV6,SIV7,(((SIV8+SIV8B)/2)+SIV9+SIV10)/3,(((SIV8+SIV8B)/2)+SIV9+SIV10)/3,(((SIV8+SIV8B)/2)+SIV9+SIV10)/3,SIV11,SIV12,na.rm=T),ifelse(((SIV8+SIV8B)/2)==SIV9 & SIV9<SIV10,min(SIV3,SIV6,SIV7,(((SIV8+SIV8B)/2)+SIV9+SIV10)/3,(((SIV8+SIV8B)/2)+SIV9+SIV10)/3,SIV10,SIV11,SIV12,na.rm=T),ifelse(((SIV8+SIV8B)/2)==SIV1 0 & SIV10<SIV9,min(SIV3,SIV6,SIV7,(((SIV8+SIV8B)/2)+SIV9+SIV10)/3,SIV9,(((SIV8+SIV8B)/2)+SIV9+SIV10)/3,SIV11,SIV12,na.rm=T),ifelse(SIV9==SIV10 & SIV9<((SIV8+SIV8B)/2),min(SIV3,SIV6,SIV7,((SIV8+SIV8B)/2),(((SIV8+SIV8B)/2)+SIV9+SIV10)/3,(((SIV8+SIV8B)/2)+SIV9+SIV10)/3,SIV11,SIV12,na.rm=T),ifelse(min(((SIV8+SIV8B)/2),SIV9,SIV10)==((SIV8+SIV8B)/2),min(SIV3,SIV6,SIV7,(((SIV8+SIV8B)/2)+SIV9+SIV10)/3,SIV9,SIV10,SIV11,SIV12,na.rm=T),ifelse(min(((SIV8+SIV8B)/2),SIV9,SIV10)==SIV9,min(SIV3,SIV6,SIV7,((SIV8+SIV8B)/2),(((SIV8+SIV8B)/2)+SIV9+SIV10)/3,SIV10,SIV11,SIV12,na.rm=T),min(SIV3,SIV6,SIV7,((SIV8+SIV8B)/2),SIV9,(((SIV8+SIV8B)/2)+SIV9+SIV10)/3,SIV11,SIV12,na.rm=T))))))))</td></tr>
<tr><td>Juvenile component</td><td>ifelse(SIV4<=0.3|SIV5<=0.3,min(SIV1,SIV2B,SIV3,SIV4,SIV5,SIV11,SIV12,SIV13,SIV14,SIV15,SIV16,SIV17,na.rm=T),ifelse(SIV4==SIV5,min(SIV1,SIV2B,SIV3,(SIV4+SIV5)/2,(SIV4+SIV5)/2,SIV11,SIV12,SIV13,SIV14,SIV15,SIV16,SIV17,na.rm=T),ifelse(SIV4<SIV5,min(SIV1,SIV2B,SIV3,(SIV4+SIV5)/2,SIV5,SIV11,SIV12,SIV13,SIV14,SIV15,SIV16,SIV17,na.rm=T),min(SIV1,SIV2B,SIV3,SIV4,(SIV4+SIV5)/2,SIV11,SIV12,SIV13,SIV14,SIV15,SIV16,SIV17,na.rm=T))))</td></tr>
<tr><td>Adult component</td><td>ifelse(SIV4<=0.3|SIV5<=0.3,min(SIV1,SIV2,SIV3,SIV4,SIV5),ifelse(SIV4==SIV5,min(SIV1,SIV2,SIV3,(SIV4+SIV5)/2,(SIV4+SIV5)/2),ifelse(SIV4<SIV5,min(SIV1,SIV2,SIV3,(SIV4+SIV5)/2,SIV5),min(SIV1,SIV2,SIV3,SIV4,(SIV4+SIV5)/2))))</td></tr>
</table>

Model equation:

The equation to calculate an overall HSI index for the chinooksalmonCompLtoe5CSilt model is:

`min(CA,CE,CJ)`

According to our classification, this model's format is: **author-specified**

Global sensitivity and uncertainty analysis:

We ran global sensitivity and uncertainty analyses on the chinooksalmonCompLtoe5CSilt model using the sensobol package in R (Puy et al. 2022). The following parameters were used for the sensobol analysis:

Table 3. Parameters and settings used for sensobol sensitivity and uncertainty analyses.

Parameter	Equation	Value
Number of input variables (M)	-	19

Base sample size (n)	-	310000
Number of model evaluations (N)	n*(M+2)	5985000
First order estimator	See Puy et al. (2022)	Saltelli
Total order estimator	See Puy et al. (2022)	Jansen
Number of bootstrap replications	-	1000
Sampling scheme	-	Quasi-random
Matrices	-	A, B, AB

We ran a sensitivity and uncertainty analysis for the chinooksalmonCompLtoe5CSilt model using the original equation outlined in the documentation from Raleigh et al. (1986) and using arithmetic mean, geometric mean, limiting factor, and multiplicative equations to contrast the results across different equation structures.

Model uncertainty

We ran the chinooksalmonCompLtoe5CSilt model using 5985000 combinations of its SIV variables, which were sampled from a uniform distribution spanning the range of possible values listed in the chinooksalmonCompLtoe5CSilt documentation. We limited the range of possible values for each parameter to the range in which the SIV values were greater than zero to prevent HSI score distributions with primarily zero values.

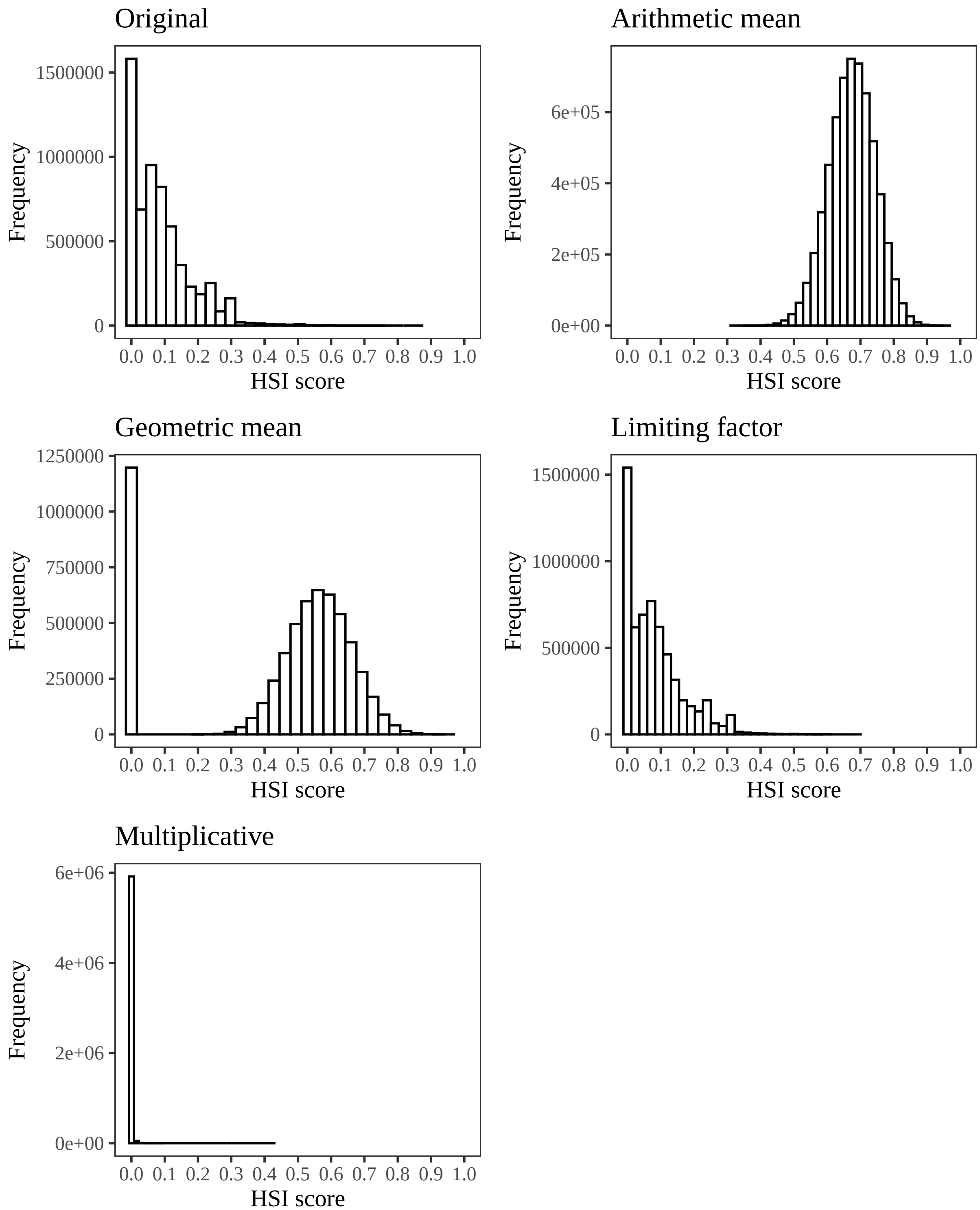

Figure 2. Empirical distributions of HSI scores for the chinooksalmonCompLtoe5CSilt model using the original author-specified model equation from Raleigh et al. (1986), and an arithmetic mean, geometric mean, limiting factor, and multiplicative structure incorporating all SIV variables. Note differences in the y axis.

We assumed a uniform distribution for all parameters because we evaluated all ecorest models in batch. Should you decide to run your own sensitivity analysis, this assumption should be evaluated independently for each parameter in the model.

Table 4. Quantiles from the empirical distribution of HSI scores for the original chinooksalmonCompLtoe5CSilt model structure, an arithmetic mean equation, a geometric mean equation, a limiting factor equation, and a multiplicative equation structure.

	1%	**2.5%**	**5%**	**25%**	**50%**	**75%**	**95%**	**97.5%**	**99%**	**100%**
Original	0.00	0.00	0.00	0.01	0.07	0.13	0.26	0.30	0.35	0.86
Arithmetic	0.51	0.53	0.56	0.63	0.68	0.72	0.79	0.81	0.83	0.96
Geometric	0.00	0.00	0.00	0.42	0.54	0.61	0.71	0.74	0.78	0.96
Limiting	0.00	0.00	0.00	0.01	0.06	0.12	0.25	0.30	0.31	0.69
Multiplicative	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.43

Generally, models with outputs that span the entire range of possible outcomes (*i.e.*, 0-1) may be better at discriminating between different restoration actions or across habitat quality. The empirical distribution of HSI scores can also provide you with information about whether HSI scores for a given model tend to be skewed (*e.g.*, many low scores and few high scores, uniform probability of obtaining scores from 0-1, *etc.*)

Model sensitivity

Below are the results of the global sensitivity analysis for the chinooksalmonCompLtoe5CSilt model using the original equation, an arithmetic mean, a geometric mean, a limiting factor, and a multiplicative model structure. The sensobol package uses variance-based sensitivity metrics, so the model's sensitivity to a given parameter is a measure of how much variance in the HSI score decreases in response to that parameter being fixed (Puy et al. 2022). For each parameter, the observed changes in the variance of the HSI score can be described with a first order sensitivity index (S_i) that accounts for the influence of a single parameter of interest on variance in HSI, or with a total order index (T_i) that accounts for the influence of a single parameter on its own and in combination with all other parameters (*i.e.*, interactions) (Puy et al. 2022). We can compare the 95% confidence intervals for the first and total order indices to a dummy parameter, which represents a parameter that has no influence on the variance in a model's output. While an uninfluential variable should theoretically have an S_i and T_i of zero, small approximation errors can lead variables to have a non-zero influence on a model's output (Puy et al. 2022). If the confidence interval of the S_i and T_i index for a given parameter overlaps the confidence interval of the dummy parameter, we can deduce that the parameter has a negligible effect on variance in HSI scores, both on its own and in combination with all other variables.

To ensure that the results of our sensitivity analysis converged, we determined whether the range of the 95% confidence interval for each sensitivity index was less than or equal to 0.05 (Sarrazin et al., 2016). Models that did not converge should be interpreted with caution.

Table 5. Model convergence records for the original chinooksalmonCompLtoe5CSilt model structure, an arithmetic mean equation, a geometric mean equation, a limiting factor equation, and a multiplicative equation structure.

	Did the model converge? (T/F)
Original	TRUE
Arithmetic	TRUE
Geometric	TRUE
Limiting	TRUE
Multiplicative	FALSE

Original model

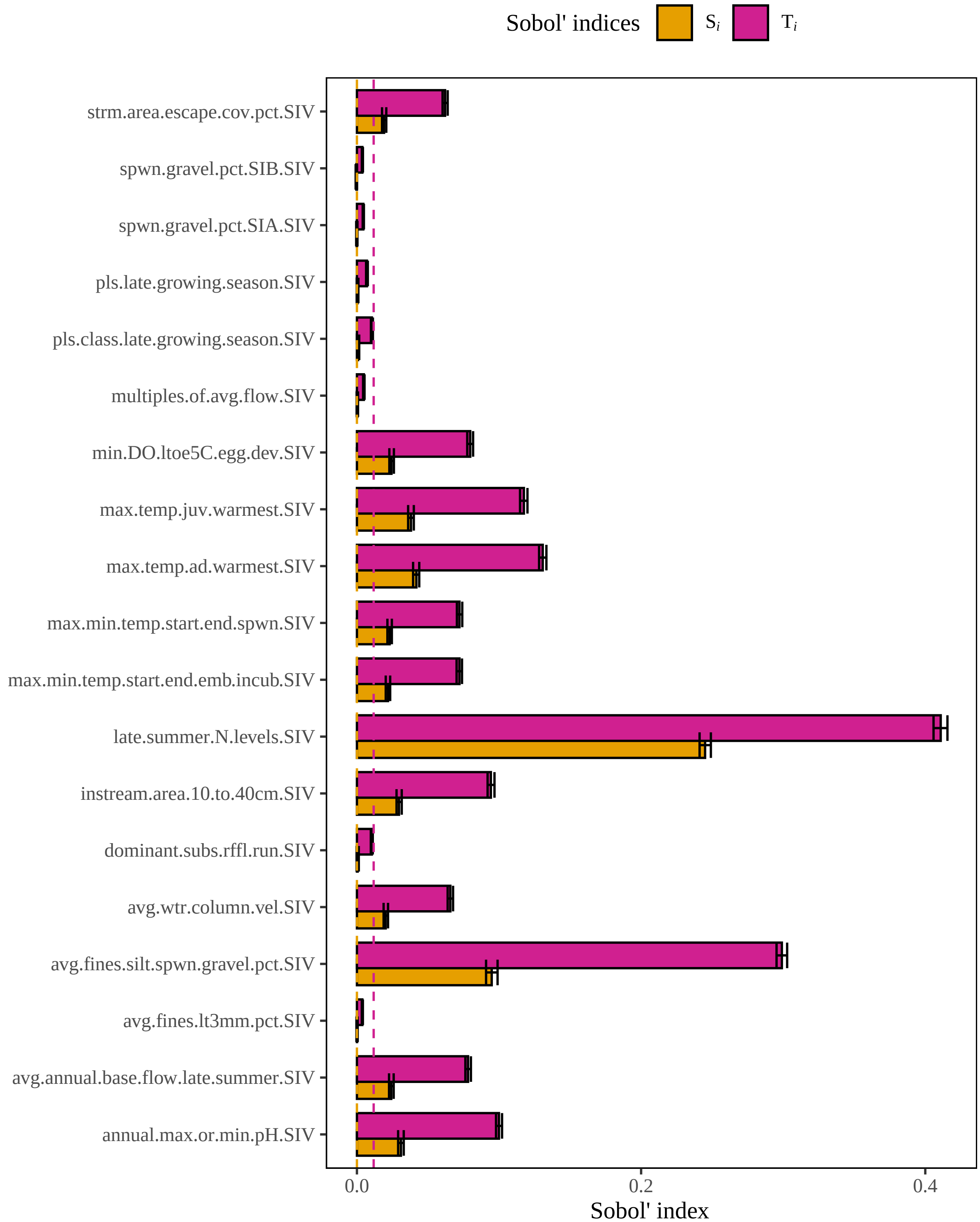

Figure 3. Results of a sensitivity analysis for the chinooksalmonCompLtoe5CSilt model based on the original author-specified model outlined in Raleigh et al. (1986). Dashed lines represent baseline numerical approximation error for S_i and T_i (*i.e.*, dummy variables).

Arithmetic mean model

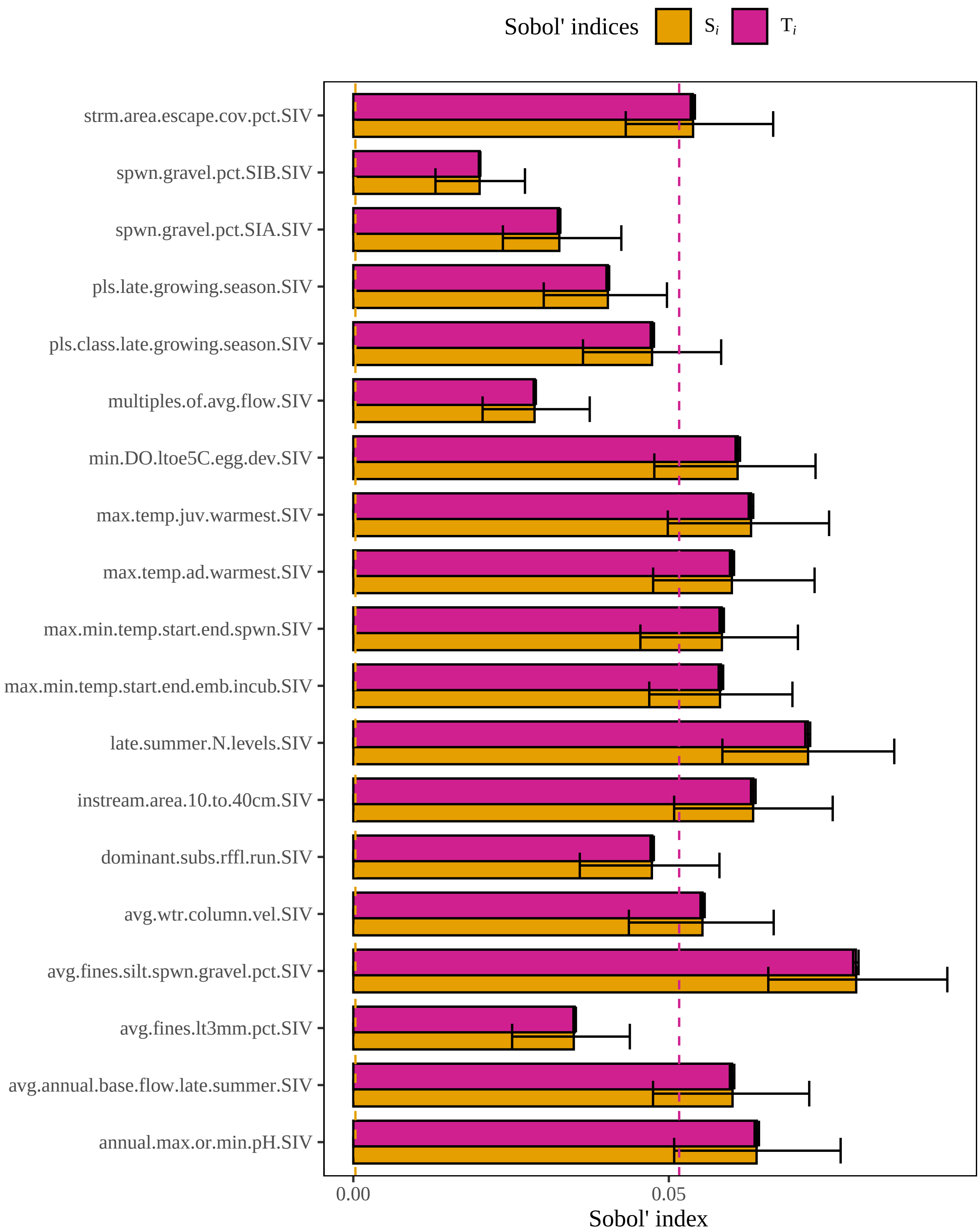

Figure 4. Results of a sensitivity analysis for the chinooksalmonCompLtoe5CSilt model based on an arithmetic mean structure. Dashed lines represent baseline numerical approximation error for S_i and T_i (*i.e.*, dummy variables).

Geometric mean model

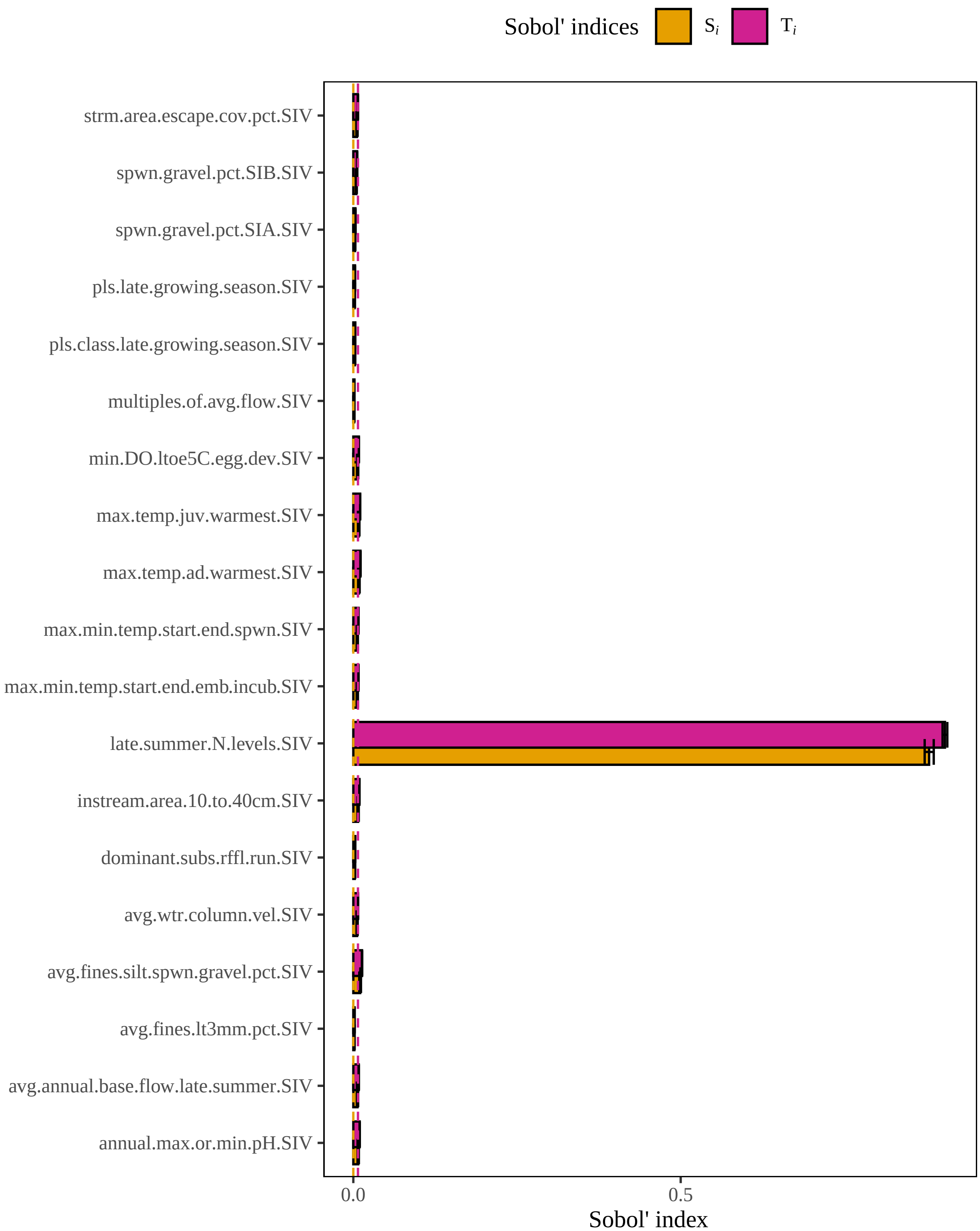

Figure 5. Results of a sensitivity analysis for the chinooksalmonCompLtoe5CSilt model based on a geometric mean structure. Dashed lines represent baseline numerical approximation error for S$_i$ and T$_i$ (*i.e.*, dummy variables).

Limiting factor model

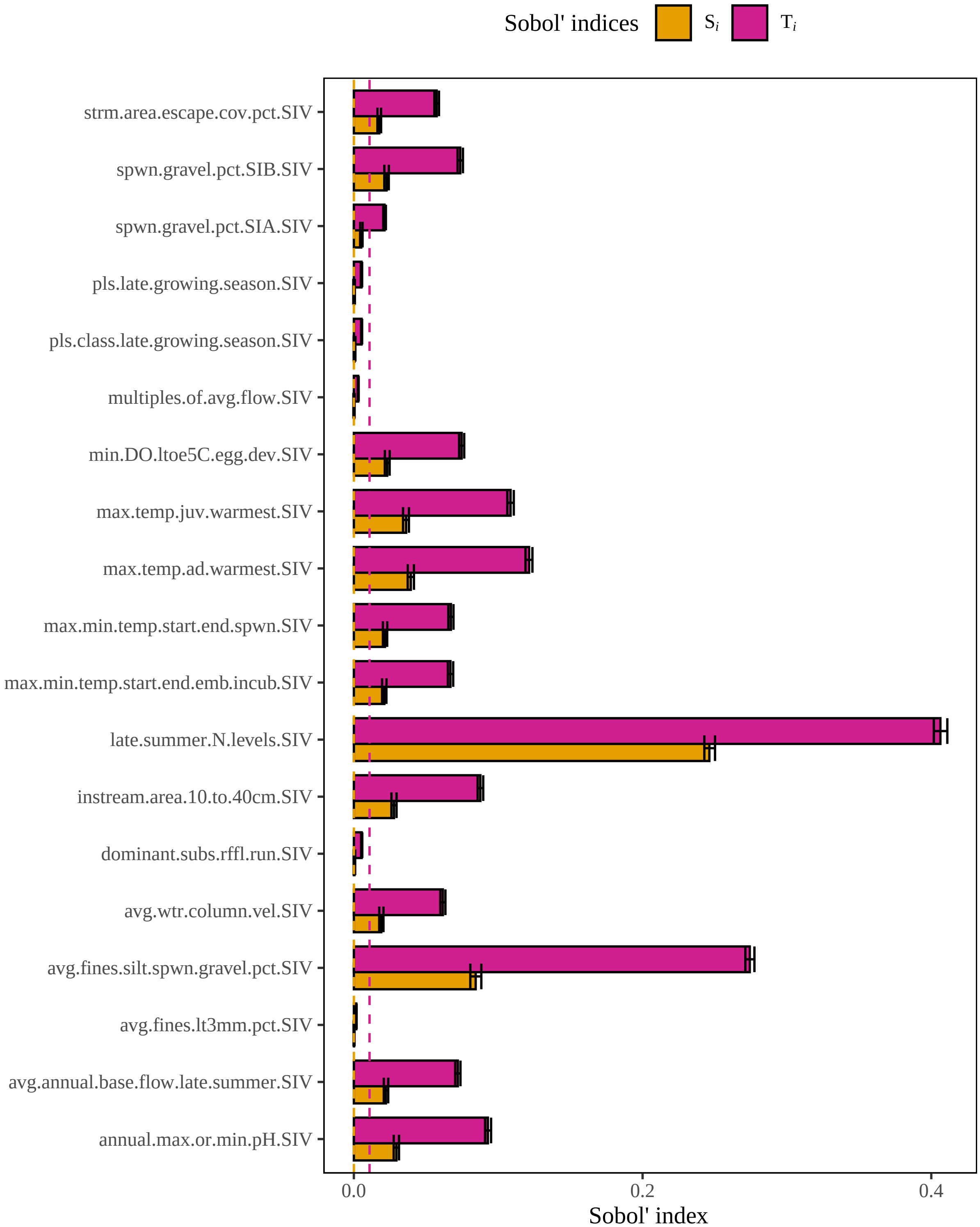

Figure 6. Results of a sensitivity analysis for the chinooksalmonCompLtoe5CSilt model based on a limiting factor structure. Dashed lines represent baseline numerical approximation error for S_i and T_i (*i.e.*, dummy variables).

Multiplicative model

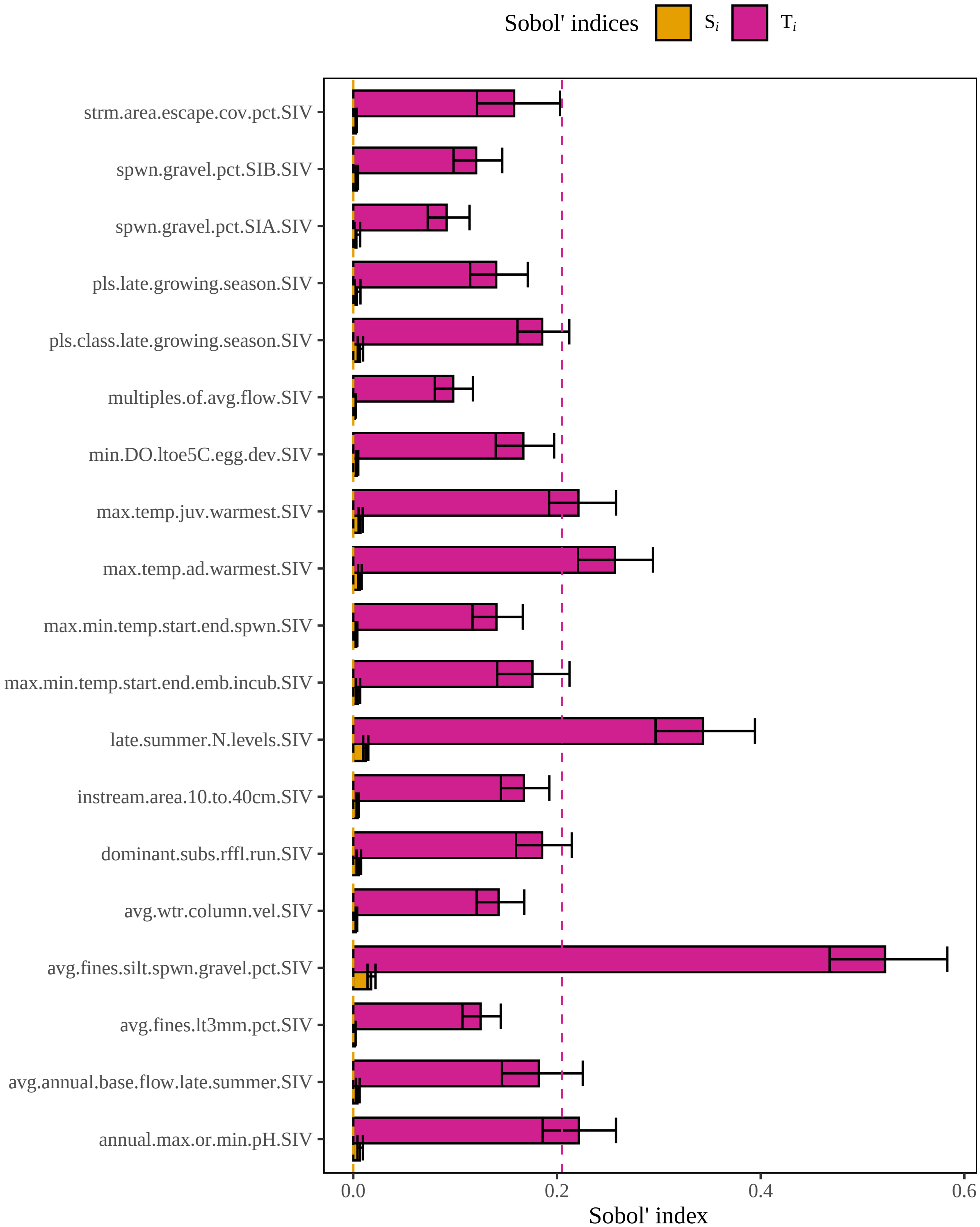

Figure 7. Results of a sensitivity analysis for the chinooksalmonCompLtoe5CSilt model based on a multiplicative mean structure. Dashed lines represent baseline numerical approximation error for S$_i$ and T$_i$ (*i.e.*, dummy variables).

Summary of influential variables

Original model

In the **original chinooksalmonCompLtoe5CSilt** model, **12 of 19** variables are influential.

Variable with the highest first order sensitivity: late.summer.N.levels.SIV

Variable with the highest total order sensitivity: late.summer.N.levels.SIV

Un-influential variables in original model:

```
pls.late.growing.season.SIV
pls.class.late.growing.season.SIV
spwn.gravel.pct.SIA.SIV
spwn.gravel.pct.SIB.SIV
multiples.of.avg.flow.SIV
dominant.subs.rffl.run.SIV
avg.fines.lt3mm.pct.SIV
```

Arithmetic mean model

In the **arithmetic mean chinooksalmonCompLtoe5CSilt** model, **12 of 19** variables are influential.

Variable with the highest first order sensitivity: avg.fines.silt.spwn.gravel.pct.SIV

Variable with the highest total order sensitivity: avg.fines.silt.spwn.gravel.pct.SIV

Un-influential variables in arithmetic mean model:

```
pls.late.growing.season.SIV
pls.class.late.growing.season.SIV
spwn.gravel.pct.SIA.SIV
spwn.gravel.pct.SIB.SIV
multiples.of.avg.flow.SIV
dominant.subs.rffl.run.SIV
avg.fines.lt3mm.pct.SIV
```

Geometric mean model

In the **geometric mean chinooksalmonCompLtoe5CSilt** model, **11 of 19** variables are influential.

Variable with the highest first order sensitivity: late.summer.N.levels.SIV

Variable with the highest total order sensitivity: late.summer.N.levels.SIV

Un-influential variables in geometric mean model:

```
pls.late.growing.season.SIV
pls.class.late.growing.season.SIV
spwn.gravel.pct.SIA.SIV
spwn.gravel.pct.SIB.SIV
multiples.of.avg.flow.SIV
dominant.subs.rffl.run.SIV
avg.fines.lt3mm.pct.SIV
strm.area.escape.cov.pct.SIV
```

Limiting factor model

In the **limiting factor chinooksalmonCompLtoe5CSilt** model, **14 of 19** variables are influential.

Variable with the highest first order sensitivity: late.summer.N.levels.SIV

Variable with the highest total order sensitivity: late.summer.N.levels.SIV

Un-influential variables in limiting factor mean model:

```
pls.late.growing.season.SIV
pls.class.late.growing.season.SIV
multiples.of.avg.flow.SIV
dominant.subs.rffl.run.SIV
avg.fines.lt3mm.pct.SIV
```

Multiplicative model

In the **multiplicative chinooksalmonCompLtoe5CSilt** model, **3 of 19** variables are influential.

Variable with the highest first order sensitivity: avg.fines.silt.spwn.gravel.pct.SIV

Variable with the highest total order sensitivity: avg.fines.silt.spwn.gravel.pct.SIV

Un-influential variables in multiplicative model:

```
annual.max.or.min.pH.SIV
max.temp.juv.warmest.SIV
min.DO.ltoe5C.egg.dev.SIV
pls.late.growing.season.SIV
pls.class.late.growing.season.SIV
max.min.temp.start.end.spwn.SIV
max.min.temp.start.end.emb.incub.SIV
spwn.gravel.pct.SIA.SIV
spwn.gravel.pct.SIB.SIV
avg.wtr.column.vel.SIV
avg.annual.base.flow.late.summer.SIV
multiples.of.avg.flow.SIV
dominant.subs.rffl.run.SIV
avg.fines.lt3mm.pct.SIV
strm.area.escape.cov.pct.SIV
instream.area.10.to.40cm.SIV
```

References

1. McKay S, D Hernandez-Abrams, and K Cushway. 2025. ecorest: conducts analyses informing ecosystem restoration decisions. R package version 2.0.1, https://CRAN.R-project.org/package=ecorest.
2. Puy, A, S Lo Piano, A Saltelli, and SA Levin. 2022. sensobol: an R package to compute variance based sensitivity indices. Journal of Statistical Software 102(5):1-37. doi: 10.18637/jss.v102.i05
3. Raleigh, RF, WJ Miller, and PC Nelson. 1986. Habitat suitability index models and instream flow suitability curves: chinook salmon. U.S. Fish Wildl. Serv. Biol. Rep. 82(10.122). 64 pp.
4. Sarrazin, F., Pianosi, F., & Wagener, T. (2016). Global sensitivity analysis of environmental models: Convergence and validation. Environmental Modelling & Software, 79, 135–152. https://doi.org/10.1016/j.envsoft.2016.02.005